Quantum Computing: Breakthroughs, Architecture, and Error Correction
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Abstract:
Quantum computing represents a paradigm shift from classical von Neumann computational models. By utilizing quantum bits (qubits) capable of existing in superpositions of basis states |0> and |1>, quantum processors achieve exponential speedups for specialized computational classes, including integer factorization, quantum chemistry simulations, and discrete optimization problems.

1. Quantum Supremacy and Coherence Time
Recent superconducting quantum processors with 128 physical qubits have demonstrated coherence times reaching 140 microseconds under cryogenic conditions of 15 millikelvin. Coherence time is strictly limited by environmental thermal noise, dielectric losses in the substrate, and stray magnetic fields. The system employs transmon qubit architecture with tunable capacitive couplers to minimize crosstalk. Benchmarking on random circuit sampling verified a linear cross-entropy benchmarking fidelity of 0.72, surpassing the computational capability of classical supercomputing clusters by an estimated factor of 10,000.

2. Quantum Error Correction and Fault Tolerance

Physical qubits are notoriously susceptible to decoherence and phase-flip errors. To realize practical, fault-tolerant quantum computation, logical qubits are synthesized using surface codes.

The distance-5 surface code requires 49 physical qubits per logical qubit to correct arbitrary single-qubit Pauli X and Z errors per syndrome cycle.

Empirical measurements indicate that the physical error threshold for surface codes is approximately 0.57% under depolarizing noise models. Our experiments demonstrated that lowering two-qubit gate error rates below 0.1% enables exponential suppression of logical error rates as code distance increases from d=3 to d=5.

The decoder uses minimum-weight perfect matching (MWPM) algorithms executing on real-time FPGA co-proce with a sub-microsecond latency budget of 420 nanoseconds.

3. Algorithmic Scaling and Cryptographic Implications
Shor's algorithm provides polynomial-time factorization of composite integers in O((log N)^3) operations,
posing an existential challenge to classical asymmetric cryptography including RSA-2048 and Elliptic Curve Diffi
Breaking RSA-2048 in under 8 hours requires an estimated 4,096 fault-tolerant logical qubits and approximately

In contrast, Grover's algorithm provides a quadratic speedup O(sqrt(N)) for unstructured database search,
reducing symmetric AES-256 security to an effective 128-bit key strength.
Consequently, post-quantum cryptographic standards based on lattice-based cryptography (CRYSTALS-Kyber a
must be adopted across enterprise infrastructure before fault-tolerant quantum hardware reaches maturity.